

BANARAS HINDU UNIVERSITY

B.Sc. (Hons) Semester IV Examination, 2025-26 (NEP)

BOTANY**Paper Code: BOTMJ41 : Gymnosperms and Paleobotany**

Ref. No.: D26(232)-300

Time: 2½ Hours

Maximum Marks: 70

Instructions: Answer five questions including Question 1 which is compulsory. The figures in the right-hand margin indicate marks.

Question 1. Answer the following questions in not more than 100 words each: $[7 \times 2 = 14]$

- (a) Why are the leaves of *Cycas* described as pinnately compound but fern-like in appearance?
- (b) Describe the salient features of Progymnosperm.
- (c) Why are the leaves of *Ephedra* reduced and what is the role of its green, jointed stems?
- (d) Give an account of the affinities of Pteridospermales with Cycadophyta and ferns.
- (e) Why are the seeds of *Pinus* winged and how are these wings derived?
- (f) Describe the anatomical features of *Lyginopteris*.
- (g) What is the significance of the presence of resin canals for survival in *Pinus*?

Question 2. Describe the geological time scale, outlining its major divisions, key events in each era and its significance in understanding the evolution of plant life. $[14]$

Question 3. Write short notes on *any two* of the following: $[2 \times 7 = 14]$

- (a) Reproductive parts of *Lyginopteris*
- (b) Pteridospermales
- (c) Classification of gymnosperms.

Question 4. Give a comprehensive account of the types of fossils and their importance, with suitable diagrams. $[14]$

Question 5. Write short notes on *any two* of the following: $[2 \times 7 = 14]$

- (a) Reproduction in *Pinus*
- (b) Difference between young and old stems of *Ephedra*
- (c) Structure and adaptations of *Pinus* leaf.

Question 6. Describe the anatomy and reproduction of *Cycas*, highlighting its distinctive features and evolutionary significance. $[14]$

Question 7. Write short notes on *any two* of the following: $[2 \times 7 = 14]$

- (a) Major geological events in Earth's history
- (b) *Glossopteris* leaf
- (c) Poikilohydry and homoiohydry.
